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corrp: An R package for multiple correlation-like analysis and clustering in mixed data

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Software

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Summary

Correlation-like analysis provides an important statistical measure that can describe the size and direction of an association between variables. However, there are few R packages that can efficiently perform this type of analysis on large datasets with mixed data types. The `corrp` package provides a full suite of solutions for computing various kinds of correlation-like measures such as Pearson correlation (Pearson, 1895), Distance Correlation (Székely et al., 2007), Maximal Information Coefficient (MIC) (Reshef et al., 2011), Predictive Power Score (PPS) (Wetschoreck et al., 2020), Cramér's V (Cramér, 1946), and the Uncertainty Coefficient (Theil, 1972). These methods support the analysis of data frames with mixed classes (integer, numeric, factor, and character). In addition, it offers a C++ implementation of the Average Correlation Clustering Algorithm (ACCA) (Bhattacharya & De, 2010), which is original used in genetic studies. In this package, the ACCA algorithm has been extended to work directly with correlation matrices derived from different association methods, depending on the data types and user preferences. Furthermore, the package is designed for parallel processing in R, making it highly efficient for large datasets.

Statement of need

The corrp package is an R package that provides a flexible and efficient way of doing correlation-like analysis on mixed-type data frames. Most traditional correlation methods in R are applicable only to specific data types or small datasets `corrp` extends this capability by handling mixed classes, integrating various association methods, and offering clustering directly from the resulting correlation matrix. The package is particularly useful for researchers and data scientists working with complex datasets who require robust and scalable tools for both association analysis and clustering.

Implementation

The `corrp` package integrates R and C++ to combine the flexibility of R with the speed of C++, optimizing key operations. Its core functionalities include the selection of correlation-like methods based on pair of variable types (numeric pairs, numeric and categorical pairs, etc.). Users can create correlation matrices, remove variables based on significance, and cluster the correlation matrix using the ACCA clustering algorithm. This approach has been modified to support mixed data types and various correlation methods. Additionally, the package supports parallel processing through the `foreach` package, significantly improving performance on large datasets.

As mentioned before, one can choose between the following options based on the type pair:

- **Numeric pairs (integer/numeric):**

  - Pearson correlation coefficient (Pearson, 1895), a widely used measure of the strength and direction of linear relationships.
  - Distance Correlation or distance covariance (Székely et al., 2007), based on the idea of expanding covariance to distances, can measure both linear and nonlinear associations between variables.
  - Maximal Information Coefficient (MIC) (Reshef et al., 2011), a information-based nonparametric based method that can detect linear or non-linear relationships between variables.
  - Predictive Power Score (PPS) (Wetschoreck et al., 2020), a metric used to assess predictive relations between variables.
- **Numeric and categorical pairs (integer/numeric - factor/categorical):**
  - Square root of the R^2 coefficient from linear regression (Cohen, 1983).
  - Predictive Power Score (PPS) (Wetschoreck et al., 2020).
- **Categorical pairs (factor/categorical):**
  - Cramér's V (Cramér, 1946), a measure of association between nominal variables.
  - Uncertainty Coefficient (Theil, 1972), a measure of nominal association between two variables.
  - Predictive Power Score (PPS) (Wetschoreck et al., 2020).

Usage

The `corrp` package provides seven main functions for correlation calculations, clustering, and basic data manipulation:

- **corrp**: Performs correlation-like analysis with user-specified methods for numeric, categorical, factor, interger and mixed pairs.
- **corr_matrix**: Generates a correlation matrix from analysis results.
- **corr_rm**: Removes variables based on p-value significance.
- **acca**: Performs the ACCA clustering algorithm with added support for mixed data types.
- **sil_acca**: A C++ implementation of the Silhouette method for interpreting and validating the consistency of clusters within ACCA clusters of data.
- **best_acca**: Determining the optimal number of clusters in ACCA clustering using the average silhouette approach.

First, we calculate the correlations for the *iris* dataset using the Maximal Information Coefficient for numeric pairs, the Predictive Power Score algorithm for numeric/categorical pairs, and the Uncertainty Coefficient for categorical pairs.

```
library("corrp")
results = corrp(
  iris,
  cor.nn = 'mic', cor.nc = 'pps', cor.cc = 'uncoef',
  n.cores = 2, verbose = FALSE
)

head(results$data)
```

	infer	infer.value	stat	stat.value
1	Maximal Information Coefficient	0.9994870	P-value	0.0000000
2	Maximal Information Coefficient	0.2770503	P-value	0.0000000
3	Maximal Information Coefficient	0.7682996	P-value	0.0000000
4	Maximal Information Coefficient	0.6683281	P-value	0.0000000
5	Predictive Power Score	0.5591864	F1_weighted	0.7028029
6	Maximal Information Coefficient	0.2770503	P-value	0.0000000

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	isig	msg	varx	vary
1	TRUE		Sepal.Length	Sepal.Length
2	TRUE		Sepal.Length	Sepal.Width
3	TRUE		Sepal.Length	Petal.Length
4	TRUE		Sepal.Length	Petal.Width
5	TRUE		Sepal.Length	Species
6	TRUE		Sepal.Width	Sepal.Length

Using the previous result, we can create a correlation matrix as follows:

```
m = corr_matrix(results, col = 'infer.value', isig = TRUE)
m
```

	Sepal.Length	Sepal.Width
Sepal.Length	0.9994870	0.2770503
Sepal.Width	0.2770503	0.9967831
Petal.Length	0.7682996	0.4391362
Petal.Width	0.6683281	0.4354146
Species	0.5591864	0.3134401

	Petal.Length	Petal.Width	Species
Sepal.Length	0.7682996	0.6683281	0.4075487
Sepal.Width	0.4391362	0.4354146	0.2012876
Petal.Length	1.0000000	0.9182958	0.7904907
Petal.Width	0.9182958	0.9995144	0.7561113
Species	0.9167580	0.9398532	0.9999758

```
# attr(,"class")
# [1] "cmatrix" "matrix"
```

Finally, we can cluster the dataset variables using the ACCA algorithm and the correlation matrix. For example, consider clustering into 2 groups (k = 2):

```
acca.res = acca(m, 2)
acca.res
# $cluster1
# [1] "Species"      "Sepal.Length" "Petal.Width"
#
# $cluster2
# [1] "Petal.Length" "Sepal.Width"
#
# attr(,"class")
# [1] "acca_list" "list"
```

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